

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 3: Mock Interview**

**COURSE NAME:** Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

**COURSE OUTCOME COVERED**

**CO5:** *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.* (BL5)

Dave's Taxonomy: Articulation (Level 4)

**Total Marks: 25**

Question Count: 10

**Code of Conduct**

I \_\_\_\_\_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: \_\_\_\_\_

**Assessment Tasks**

**Mock Interview**

Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?
9. How would you design a secure data ingestion pipeline for a large organization?
10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

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### 1. Explain the difference between HDFS and traditional file systems.

HDFS (Hadoop Distributed File System) is designed to store very large datasets across multiple machines. It divides files into blocks and distributes them across DataNodes, with replication for fault tolerance.

A traditional file system usually stores files on a single machine or centralized storage system.

#### **Example:**

Traditional: File → One Server

HDFS: File → Blocks → Multiple DataNodes

**Key difference:** HDFS provides distributed storage, scalability, and fault tolerance, while traditional file systems mainly focus on storage within a single system.

### 2. Why is replication important in distributed storage systems?

Replication means maintaining **multiple copies of the same data** on different nodes.

It is important because if one node fails, another copy can still provide the data.

**For example, HDFS commonly uses a replication factor of 3:**

#### **Block A**

|— DataNode 1

|— DataNode 2

|— DataNode 3

If DataNode 1 fails, DataNode 2 or 3 can provide the block.

Therefore, replication improves availability, fault tolerance, and reliability.

### 3. How does MapReduce divide work between map and reduce phases?

MapReduce divides large processing tasks into two main phases.

**Map phase:** Reads input data and produces intermediate key-value pairs.

**Reduce phase:** Groups and combines the intermediate values to generate the final result.

Input → Mapper → Shuffle & Sort → Reducer → Output

For example, in word counting:

Mapper: Hadoop → 1

Reducer: Hadoop → 1000

This allows processing to happen in parallel across multiple nodes

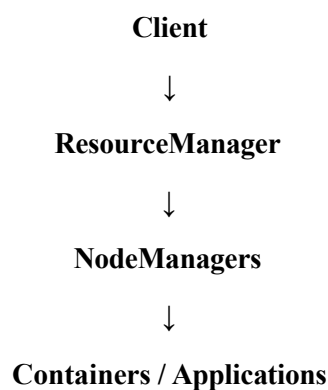
### 4. What role does YARN play in large-scale Hadoop processing?

YARN (Yet Another Resource Negotiator) is Hadoop's resource management and job scheduling layer.

It manages:

- CPU and memory resources.
- Job scheduling.
- Application execution.
- Cluster resource allocation.

**Architecture:**



YARN allows multiple applications such as **MapReduce**, **Spark**, **ETL**, and **machine-learning jobs** to share the same Hadoop cluster efficiently.

## 5. Differentiate NAS and SAN in an enterprise data environment.

| NAS                                           | SAN                                                      |
|-----------------------------------------------|----------------------------------------------------------|
| Network Attached Storage                      | Storage Area Network                                     |
| Provides file-level access                    | Provides block-level access                              |
| Usually accessed through network protocols    | Usually uses high-speed storage networks                 |
| Easier to manage                              | More complex                                             |
| Suitable for file sharing and general storage | Suitable for databases and high-performance applications |

### Simple example:

NAS → Users/Applications → Shared Files

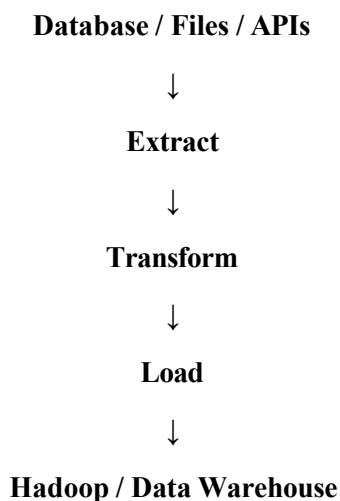
SAN → Servers → High-performance Block Storage

So, NAS is mainly file-oriented, while SAN is block-oriented.

## 6. What is the purpose of ETL in a big data pipeline?

ETL stands for:

- **Extract** – collect data from different sources.
- **Transform** – clean, validate, and convert the data.
- **Load** – store the processed data in the target system.



ETL ensures that data is clean, consistent, and ready for analytics.

## 7. How does Apache NiFi help in data integration workflows?

**Apache NiFi** is a data-flow and integration tool used to move and process data between different systems.

It can:

- Collect data from multiple sources.
- Route and filter data.
- Transform data.
- Transfer data to HDFS or databases.
- Monitor data flow.

**Example:**

**Database + NAS + APIs**



**Apache NiFi**



**HDFS/Data Lake**



**Analytics**

Its main advantage is that it provides visual, manageable, and traceable data pipelines.

## 8. What are the key failure points in a Hadoop cluster and how would you handle them?

Important failure points include:

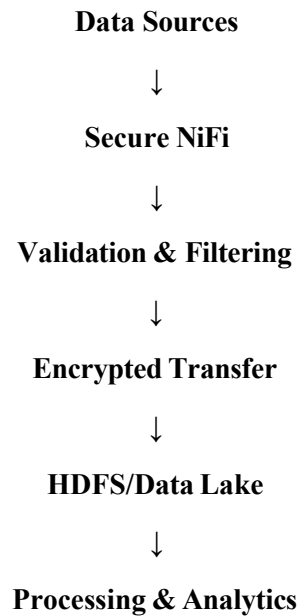
1. **DataNode failure** – handled using HDFS replication and automatic re-replication.
2. **NameNode failure** – handled using Active/Standby NameNode High Availability.
3. **ResourceManager failure** – handled using YARN ResourceManager HA.
4. **Network failure** – handled through redundant network paths and monitoring.
5. **Task failure** – Hadoop can reschedule failed tasks on healthy nodes.

Architecture:

Failure → Detection → Recovery/Rescheduling

The main principle is to **avoid single points of failure**.

## 9. How would you design a secure data ingestion pipeline for a large organization?



Security mechanisms would include:

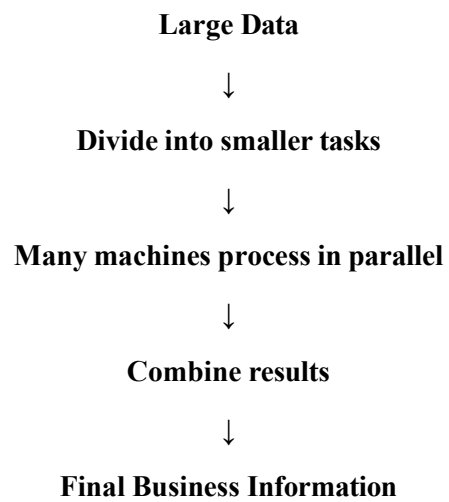
- **TLS/HTTPS** for data in transit.
- Authentication and authorization.
- Role-based access control.
- Encryption for sensitive data.
- Data validation.
- Audit logs and data provenance.
- Secure credential management.

This ensures data is secure, traceable, and protected during movement and storage.

## 10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Think of Hadoop as a large team of workers. Instead of asking one worker to process a huge amount of data, Hadoop divides the work among many workers and combines their results. When the amount of data increases, we can simply add more workers. If one worker fails, another worker can continue the task using a copy of the data.

Architecture-level explanation:



Therefore, Hadoop provides scalability, parallel processing, fault tolerance, and cost-efficient expansion.

### Mock Interview Rubric

| Criteria                       | Weightage | Level 4 - Exemplary                                 | Level 3 - Proficient         | Level 2 - Developing        | Level 1 - Beginning                |
|--------------------------------|-----------|-----------------------------------------------------|------------------------------|-----------------------------|------------------------------------|
| Technical Knowledge            | 25%       | Shows deep and accurate technical understanding     | Good technical understanding | Basic understanding         | Weak understanding                 |
| Problem Solving                | 20%       | Responds with strong architecture-level reasoning   | Reasonable reasoning         | Basic reasoning             | Weak reasoning                     |
| Communication                  | 20%       | Answers are confident, precise, and professional    | Mostly clear communication   | Moderate clarity            | Weak communication                 |
| Practical Awareness            | 20%       | Connects concepts to real-world implementation well | Some practical relevance     | Limited practical relevance | No practical linkage               |
| Confidence and Professionalism | 15%       | Highly confident and professional                   | Generally professional       | Moderately professional     | Low confidence and professionalism |











